

# OSNIAS-CLEARING

## Funding & Technical Roadmap — up to the security audit

Version 0.6 | October 2026 | Distribution version — milestone-funded integrated PoC

### OBJECTIVE

Structure the next phase of Osnias Clearing into fundable technical building blocks, building on the foundation already in place — Books I–VII documentation, testnet deployments, regulatory work and ecosystem contacts — up to the integrated demonstrator, security hardening and independent audit.

## Overview

| Existing foundation                                                                                                              | PACKAGE 1 —<br>Jurisdictional Node Stack                                                                                                         | PACKAGE 2 — Common<br>Clearing Core                                                                                                                                                                     | PACKAGES 3–4 —<br>Validation & audit                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Architecture and Books I–VII<br>Sei / Sepolia testnet drafts<br>AMF / AFA / CSSF pre-<br>assessment work<br>Exploratory contacts | KYB/KYC/AML onboarding<br>SQL + API / backend<br>0x accounts + Osnias-ID<br>Node Registrar<br>EVM escrow / controller<br>Node-operator interface | Netting engine + canonical<br>ledger<br>3 chambers / currency: 1W, 4W,<br>13W<br>Chamber end-user interfaces<br>Time + solvency oracles<br>ISO 20022 + reconciliation<br>Sei adapter / O-POP interfaces | PACKAGE 3: multi-node<br>integration, stress tests,<br>demonstration<br>PACKAGE 4: audit readiness,<br>independent audit, remediation /<br>re-test |

**KEY EXISTING FOUNDATION:** assets available before funding | **PACKAGES 1–2:** development of the two software modules | **PACKAGE 3:** integration, stress tests & demonstration | **PACKAGE 4:** independent security audit.

## Funding principle

The objective is to structure Packages 1 to 4 as one integrated technical PoC, funded progressively by milestones. Each tranche funds a verifiable deliverable; acceptance of a milestone unlocks the next phase. TRLs, eligible expenditure, aid intensity and funding schemes remain to be qualified with the funding partner.

## Founder, market and regulatory framework

### Project founder

Osnias Clearing was conceived, designed, organised and documented by its founder, Denis Bouzon, with 100 % self-funding to date. The project has now reached the stage where it needs support to complement the founder's skills for the next steps: clean code rework, security, the multi-node demonstrator and audit.

### Market: a solution that does not yet exist

Osnias offers companies a netting solution that does not exist today. Multilateral netting is a mechanism proven for decades, but reserved for interbank and market infrastructures: as an order of magnitude, CLS and Clearstream together process around EUR 4,300 billion of notional volume per day, i.e. more than EUR 1 million billion per year. Recurring commercial obligations between companies have no equivalent. The aim is therefore not to take a share of an existing market, but to open this mechanism to inter-company flows.

### Regulatory framework and risks

The main risk is regulatory: qualification of the infrastructure and of the clearing token. Pre-assessment is under discussion with the AMF; the files submitted to the CSSF (Luxembourg) and the AFA (Andorra) are awaiting discussion. The sector is waiting for this pre-assessment before any commitment: until then, exchanges with potential node operators remain exploratory.

The following design principles limit risk by construction (details: [osnias-clearing.com/faq/](https://osnias-clearing.com/faq/)):

| Principle                               | Scope                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Non-custodial</b>                    | Osnias holds neither user funds nor private keys or seed phrases; collateral is escrowed separately on Ethereum.             |
| <b>No credit, overdraft or leverage</b> | Execution invariant: instantaneous net exposure ≤ available collateral; any obligation exceeding it is rejected by the code. |
| <b>Individualised collateral</b>        | No common guarantee fund: risk stays with each participant.                                                                  |

| Principle                    | Scope                                                                                                                         |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>No central novation</b>   | Osnias is not designed as a CCP; distributed multilateral netting, final qualification to be determined with the authorities. |
| <b>Node responsibilities</b> | Admission, blocking and exit of participants are decided by the node manager, under its local framework.                      |
| <b>No PFMI claim</b>         | No PFMI certification or compliance is claimed; the architecture refers to it for risk management.                            |

*Founder's estimate: around EUR 215k over 6 months for Packages 1 to 4, excluding the regulatory Package 5, physical O-POP infrastructure, mainnet and operations.*

## Foundation already in place before the funding phase

The project is not starting from scratch. The work already completed forms the foundation from which Packages 1 to 4 of the PoC can be qualified, budgeted and matched to suitable funding schemes.

| Area                                 | Elements already in place                                                                                                                                                                                                                                                                                                                                                                                                                  | Status / scope                                                                                   |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Technical documentation</b>       | Structured Books I–VII technical corpus covering the architecture, Osnias-ID, clearing cycles, security and rules, onboarding and the O-POP / RPC architecture. Related position papers and public documentation.                                                                                                                                                                                                                          | Documentation baseline available; to be kept aligned with the technical baseline.                |
| <b>Draft testnet development</b>     | Eight working draft components: 1W time oracle, Sei registry (identity), Sei controller and ORUSD ledger on Sei Atlantic-2 (Sei contracts verified); Node Registrar, EVM registry, EVM escrow and controller on Ethereum Sepolia; ORUSD Clearing Desk platform online. Details: <a href="https://osnias-clearing.com/deployment/">osnias-clearing.com/deployment/</a> and <a href="https://osnias-clearing.com/end-user/">/end-user/</a> . | Draft development on testnet; code to be reworked (Cursor AI → Foundry → Slither) before audit.  |
| <b>Regulatory pre-assessment</b>     | France — AMF: memorandum and regulatory pre-assessment file. Andorra — AFA: bilingual pre-assessment file submitted. Luxembourg — CSSF: bilingual pre-assessment file submitted.                                                                                                                                                                                                                                                           | Preparatory regulatory work already completed across three jurisdictions.                        |
| <b>Ecosystem &amp; third parties</b> | Exploratory contacts with banking, fintech and infrastructure / technology players, as well as other institutional stakeholders.                                                                                                                                                                                                                                                                                                           | Exploratory contacts and exchanges only; no partnership or consortium is presented as committed. |
| <b>Outreach &amp; traceability</b>   | Public website, summary FAQ, FR / EN documentation, architecture diagrams and technical publications enabling a third party to assess the project quickly before accessing detailed documents.                                                                                                                                                                                                                                             | Qualification material for funders, technical partners and authorities.                          |

*For a funder: the protocol already has working testnet components and a complete technical corpus. The phase to be funded covers two distinct software modules — the jurisdictional Node Stack followed by the common Clearing Core — then integration/stress testing and the independent security audit.*

## Components already developed in draft (testnet)

The components below are deployed on testnet in draft development: they will be reworked through a clean development chain before the baseline is frozen and audited.

| Component                      | Network        | Status             | What works                                                                        |
|--------------------------------|----------------|--------------------|-----------------------------------------------------------------------------------|
| <b>1W time oracle</b>          | Sei Atlantic-2 | Deployed, verified | Weekly cycle, MINT / BURN / CLEARING / P2P windows, cycle number and start block. |
| <b>Sei registry (identity)</b> | Sei Atlantic-2 | Deployed, verified | Identity of participants and nodes on the clearing side.                          |
| <b>Sei controller</b>          | Sei Atlantic-2 | Deployed, verified | Links oracle, registry, clearing token and the Sepolia EVM chain.                 |
| <b>ORUSD clearing ledger</b>   | Sei Atlantic-2 | Deployed           | USD ledger positions; MINT, BURN, P2P and transfer instructions.                  |

| Component                                | Network          | Status        | What works                                                                                   |
|------------------------------------------|------------------|---------------|----------------------------------------------------------------------------------------------|
| <b>Node Registrar</b>                    | Ethereum Sepolia | Deployed      | Registration of KK-NNN nodes (first node: 90-017).                                           |
| <b>EVM registry</b>                      | Ethereum Sepolia | Deployed      | KK zone registry on the EVM side.                                                            |
| <b>EVM escrow &amp; controller</b>       | Ethereum Sepolia | Deployed      | Collateral escrow and control on the EVM side.                                               |
| <b>Clearing platform (Clearing Desk)</b> | Sei Atlantic-2   | Online, draft | Wallet connection, ledger position, oracle state, function availability by window, registry. |

Addresses, creation transactions and verification parameters: [osnias-clearing.com/deployment/](https://osnias-clearing.com/deployment/) · platform: [osnias-clearing.com/end-user/](https://osnias-clearing.com/end-user/)

## Clean rework chain for the draft code

The tool matters less than the framing: each component is reworked in strict conformity with the technical specifications of Books I to VII.

- Cursor AI — assisted rewrite of the draft code from frozen specifications, consistent conventions, NatSpec documentation.
- Foundry — unit tests, fuzzing, invariant tests and multi-participant simulations.
- Slither — static analysis, detector review and remediation, dependency review.
- Verified testnet redeployment, then freeze of the baseline handed over to security hardening and audit.

Rework chain: from draft testnet to an auditable baseline



Figure 1 — Rework chain: from draft testnet to an auditable baseline

## Documentation corpus — Books I to VII

The technical corpus is published as seven books, each available in French and English on the public website and archived on Zenodo with a permanent DOI.

| Book       | Title                                      | Release | Public page                                                                               | Zenodo DOI                                                                    |
|------------|--------------------------------------------|---------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>I</b>   | <b>Osnias-ID</b>                           | 3.0     | <a href="https://osnias-clearing.com/osniasid/">osnias-clearing.com/osniasid/</a>         | <a href="https://doi.org/10.5281/zenodo.22978105">10.5281/zenodo.22978105</a> |
| <b>II</b>  | <b>Architecture</b>                        | 2.0     | <a href="https://osnias-clearing.com/architecture/">osnias-clearing.com/architecture/</a> | <a href="https://doi.org/10.5281/zenodo.23012840">10.5281/zenodo.23012840</a> |
| <b>III</b> | <b>Clearing rules</b>                      | 1.1     | <a href="https://osnias-clearing.com/rules/">osnias-clearing.com/rules/</a>               | <a href="https://doi.org/10.5281/zenodo.22922199">10.5281/zenodo.22922199</a> |
| <b>IV</b>  | <b>Clearing cycles 1W / 4W / 13W</b>       | —       | <a href="https://osnias-clearing.com/cycle/">osnias-clearing.com/cycle/</a>               | <a href="https://doi.org/10.5281/zenodo.22979580">10.5281/zenodo.22979580</a> |
| <b>V</b>   | <b>Security topology &amp; resilience</b>  | 2.0     | <a href="https://osnias-clearing.com/security/">osnias-clearing.com/security/</a>         | <a href="https://doi.org/10.5281/zenodo.22981575">10.5281/zenodo.22981575</a> |
| <b>VI</b>  | <b>Onboarding via a node</b>               | 1.0     | <a href="https://osnias-clearing.com/onboarding/">osnias-clearing.com/onboarding/</a>     | <a href="https://doi.org/10.5281/zenodo.23010805">10.5281/zenodo.23010805</a> |
| <b>VII</b> | <b>O-POP Osnias — mainnet architecture</b> | 1.0     | <a href="https://osnias-clearing.com/pop/">osnias-clearing.com/pop/</a>                   | <a href="https://doi.org/10.5281/zenodo.23055994">10.5281/zenodo.23055994</a> |

Each public page offers the FR / EN versions and the book PDFs. Releases as at the distribution date of this document.

## PoC architecture — two software modules, then validation

The PoC is structured around two distinct software products. The Node Stack is jurisdictional and carries onboarding, local compliance and the node's EVM functions. The Clearing Core is transversal and common to all jurisdictions; Sei remains the reference deployment, with network-specific dependencies isolated in an adapter.

### Separation principle

The Node Stack decides who is admitted and applies the local framework. The Clearing Core runs clearing once the participant is admitted. Package 3 stress-tests the integrated system; Package 4 audits it.

## PACKAGE 1 — Jurisdictional Node Stack — Ethereum / EVM

### TO DEVELOP ON THE EXISTING EVM FOUNDATION — JURISDICTION-SPECIFIC

Purpose: provide an operational testnet node able to admit a company, create its identities and technical accounts, register its Osnias-ID and operate the required EVM functions under the applicable local rules.

- Corporate onboarding: KYB/KYC/AML, representatives, beneficial owners, admission status and evidence records.
- Business backend: SQL database, API / PHP modules or equivalent, onboarding state management and audit trail.
- Provisioning of 0x accounts and mapping company ↔ Osnias-ID ↔ node ↔ currency ↔ cycle.
- Node Registrar, EVM registry, escrow and controller on Ethereum testnet; entry/exit controls and events required by clearing.
- Node-operator interface: admission, suspension, exit, collateral, alerts and operational status.
- Provisioning of the VPN account / identity for the future O-POP; physical O-POP deployment remains outside the pre-audit PoC.
- Local conversion / settlement interfaces prepared as integration points, without imposing one regulatory model across jurisdictions.

#### Milestone 1 — exit criterion

A reference jurisdictional Node Stack works end-to-end on testnet: onboarding → 0x accounts → Osnias-ID → EVM, with logs and node-operator interface.

## PACKAGE 2 — Common Clearing Core — Sei reference deployment

### EXISTING TESTNET DRAFTS — CLEAN REWORK, INTEGRATION AND EXTENSION

Purpose: build the common transversal clearing structure, independent of node-local law and replicable for each currency.

- Multilateral netting engine, canonical clearing ledger and consistency invariants.
- Three chambers per currency: 1W, 4W and 13W, each with cycle states, MINT / BURN / P2P / CLEAR rules and a dedicated end-user interface.
- 1W / 4W / 13W time oracles and clearing-side solvency oracle; redundancy, freeze and recovery policies.
- Clearing-side Osnias-ID and mapping to admitted nodes.
- Clearing-side ISO 20022: mapping of Osnias events, orders, confirmations, statuses and error handling.
- Sei ↔ Ethereum cross-layer reconciliation, divergence detection and fail-closed behaviour.
- Core built on standard EVM primitives with a Sei adapter; future portability and migration / multi-deployment without rewriting business logic.
- Clearing-side O-POP interfaces and specifications; physical O-POP installation takes place after Package 5, during industrialisation.

#### Milestone 2 — exit criterion

The common Clearing Core runs all three 1W / 4W / 13W horizons on testnet, with chamber interface, netting, oracles, ISO 20022 and reconciliation operational.

## PACKAGE 3 — Integration, Stress Tests & Demonstration

### SYSTEM VALIDATION — NOT A THIRD BUSINESS MODULE

Purpose: connect several Node Stacks to the Clearing Core, run complete cycles and expose the system to nominal, load and failure scenarios before audit.

- Demonstrator with at least three nodes and several simulated participants.
- Complete 1W / 4W / 13W cycles and flow datasets of different sizes.
- Load stress: number of obligations, event frequency, state changes and cycle windows.
- Adversarial scenarios: insufficient collateral, suspended participant, invalid messages, out-of-window operations.
- Sei / Ethereum desynchronisation, delayed or unavailable oracle, connection loss, latency and recovery.
- Systematic validation of fail-closed behaviour, logs, evidence and recovery procedures.
- Measurement of the netting rate and production of reproducible metrics for the audit package.

#### Milestone 3 — exit criterion

Several multi-node cycles are executed and documented; failure and stress scenarios are reproducible; no known critical issue remains open before audit.

## PACKAGE 4 — Security Audit

### AUDIT READINESS → INDEPENDENT AUDIT → REMEDIATION → RE-TEST

- Freeze of the code and reference baseline.
- Audit package: architecture, specifications, invariant list, threat model, tests, coverage, procedures and Package 3 evidence.
- Final security hardening: static analysis, fuzzing, dependencies, access controls, DoS and attack surfaces.

- Independent audit of smart contracts and critical components within the agreed scope.
- Remediation plan, fixes and auditor re-test.

#### Milestone 4 — exit criterion

Independent audit report delivered; vulnerabilities addressed; remediation re-tested; audited baseline frozen for the complete regulatory filing.

## Timeline, budget and milestone-based funding

Founder's estimate for the integrated PoC: EUR 215k over approximately 6 months. Amounts are indicative and must be refined against eligible expenditure, aid intensity and selected co-funding.

| Package                              | Period | Indicative budget | Milestone / deliverable                                                                      | Unlocks next phase                         |
|--------------------------------------|--------|-------------------|----------------------------------------------------------------------------------------------|--------------------------------------------|
| <b>PACKAGE 1 — Node Stack</b>        | M1–M2  | EUR 60k           | Jurisdictional testnet node: onboarding → Osnias-ID → EVM, node-operator interface and logs. | M1 acceptance → Package 2 funding          |
| <b>PACKAGE 2 — Clearing Core</b>     | M2–M4  | EUR 70k           | Common core: netting, 3 chambers / currency, end-user, oracles, ISO 20022, reconciliation.   | M2 acceptance → Package 3 funding          |
| <b>PACKAGE 3 — Stress &amp; Demo</b> | M4–M5  | EUR 45k           | Multi-node demonstrator, complete cycles, load, failures, metrics and netting rate.          | M3 acceptance → Package 4 funding          |
| <b>PACKAGE 4 — Security Audit</b>    | M5–M6  | EUR 40k           | Frozen baseline, independent audit, remediation and re-test.                                 | M4 acceptance → opens regulatory Package 5 |

*Total Packages 1 to 4: EUR 215k. The exact payment profile within each tranche (advance, interim payment, acceptance balance) will be defined with the funder(s).*

## PoC funding architecture

A milestone may be funded by one or several contributors: public grant, private co-funding or industrial partner. A contributor may participate in one tranche or several, allowing the PoC to be funded progressively without requiring a single funder.

For strategic private co-funders, a later equity investment in the future operating company may be negotiated separately. PoC funding and equity investment remain distinct transactions; participation in the PoC does not automatically confer equity rights.

## After the PoC: regulatory filing, O-POP and mainnet

| Stage                                                     | Purpose                                                                                                                     | Position in pathway                                                                                                                                      | Budget / timeline                                                                   |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>PACKAGE 5 — Complete pre-mainnet regulatory filing</b> | Submit the complete file with audited baseline, stress-test results, procedures and security evidence.                      | Continuation of the AMF pre-assessment already under way: the authority reviews a known and pre-assessed project, now supported by its audited baseline. | To determine after pre-assessment feedback: applicant, adviser, procedure and cost. |
| <b>PACKAGE 6 — O-POP &amp; industrialisation</b>          | Physical deployment of the clearing-side O-POP: servers/private RPC, VPN, DMZ, logs, monitoring, hardening and acceptance.  | After pre-mainnet regulatory framing; deployed jurisdiction by jurisdiction.                                                                             | Outside Packages 1–4 PoC; to be costed per jurisdiction.                            |
| <b>PACKAGE 7 — Mainnet</b>                                | Production deployment of the Clearing Core on reference Sei and EVM components on Ethereum; activation of authorised nodes. | After Package 5 and O-POP industrialisation. The core remains chain-agnostic and portable.                                                               | Outside PoC; deployment and operating budget to be costed.                          |

## Expected work with an innovation funding partner

- Qualify entry / exit TRL for each Package, without imposing one TRL on the whole project.
- Refine indicative budgets against genuinely eligible expenditure and aid intensity.
- Identify French, European or local schemes suited to each milestone.
- Structure tranche sequencing and acceptance / release conditions.
- Identify useful co-funders or technical partners; allow an actor to fund a specific milestone.
- Where relevant, prepare the separate articulation with strategic investors in the future operating company.
- Prepare the post-audit transition to regulatory Package 5, then O-POP industrialisation and mainnet.

### Purpose

Fund one integrated PoC through verifiable milestones: jurisdictional Node Stack → common Clearing Core → integration & stress tests → audit; then submit the complete regulatory file for a project already pre-assessed before O-POP industrialisation and mainnet.

# OSNIAS CLEARING

## Funding & Demonstrator — Executive One-Pager

Executive summary | October 2026 | Integrated milestone-funded PoC

### STARTING POINT

The project already has documented architecture, a Books I–VII corpus, working components on Sei / Sepolia and regulatory pre-assessment under way. Funding is sought to move to two clean integrated software modules, stress-validate them and obtain an independent audit.

## 1. PoC architecture

| Package | Module / phase                                | Budget  | Milestone                                                           |
|---------|-----------------------------------------------|---------|---------------------------------------------------------------------|
| 1       | Jurisdictional Node Stack —<br>Ethereum / EVM | EUR 60k | Complete testnet node:<br>onboarding → Osnias-ID → EVM              |
| 2       | Common Clearing Core —<br>reference Sei       | EUR 70k | 3 chambers / currency + netting +<br>oracles + ISO 20022 + end-user |
| 3       | Integration, Stress Tests &<br>Demonstration  | EUR 45k | Multi-node, load, failures, metrics,<br>netting rate                |
| 4       | Security Audit                                | EUR 40k | Audit + remediation + re-test;<br>frozen baseline                   |

*Indicative Packages 1–4 budget: EUR 215k / ~6 months. Each accepted milestone unlocks the next tranche; one or several funders may contribute to a given tranche.*

## 2. After the audit

- Package 5: complete pre-mainnet regulatory filing, continuing the AMF pre-assessment already under way.
- Package 6: physical clearing-side O-POP and industrialisation, jurisdiction by jurisdiction.
- Package 7: reference Sei + Ethereum mainnet, then operation.

## 3. Funding model

Possible mix of grants and private co-funding, milestone by milestone. Strategic private partners may, under a separate agreement, consider an equity investment in the future operating company; this remains distinct from PoC funding.

### OBJECTIVE

Move Osnias from an already documented testnet foundation to an integrated, stress-tested and audited baseline, then to the complete pre-mainnet regulatory filing.